ItaSoRL

RESEARCH EDITION

READ THIS FIRST · COVER PAGE ADDED 10 OCTOBER 2026

This walkthrough is the July 2026 record. Parts of it have since been corrected or withdrawn.

The pages after this one are kept exactly as they were printed in July 2026, before the September 2026 boundary checks and the October 2026 trainer correction. An October review found a bug in how the survival trainer valued the last step of an episode; the trainer was fixed and the primary runs were rerun. The numbers on these pages come from the original trainer, so read them as history. The current record is `docs/CORRECTIONS.md` and section 17 of `docs/FINDINGS.md` in the repository, github.com/iLevyTate/ItaSoRL.

WHAT HAS CHANGED SINCE THESE PAGES WERE PRINTED

1. The headline. These pages lead with 0.752 from the original trainer. On the corrected trainer, agents trained to survive and to predict their next observation read **0.733** (t-based 90% CI 0.669 to 0.797, n = 10). The registered rule is **not met**: its check with both worlds authentic (the L0 gate) does not pass on the registered pair of evaluation worlds (0.559), and on eight independent pairs it accepts on two. The three decodability clauses are met. FINDINGS 17.2, 17.5.

2. "Survival-specific" is withdrawn wherever these pages use it, the small L2 trace included. Fed the exact senses the surviving agent lived through, a brain trained only to predict reads the world within about 0.01 of the survival brain (0.721 against 0.733). The signal rides on where the foraging agent goes, and a brain trained to predict keeps it. Without the prediction task the survival agent reads 0.613, under the bar. FINDINGS 17.4, 17.6.

3. Behavior subtracted. The 0.726 these pages report with behavior subtracted is what one linear control over four behavior channels left on the original agents. On the corrected agents the best-fitting nonlinear control over observations, actions and behavior leaves 0.620, under the bar. FINDINGS 17.8.

4. A footprint, not a kept belief. The common-garden signal (0.666 forward, 0.684 reverse) is carried mostly by the footprint the prefix leaves in the body and the world. Under identical input the prefix condition fades within a few steps, and with the recurrent state zeroed the footprint still reads 0.662. FINDINGS 17.7.

5. The 0.99 against 0.51 contrast. At the detector noise the L3 gate uses, the same detector reads the L2 flaws at 0.62 to 0.65, so the matched contrast is L1: detector 0.873, survival agent 0.533. FINDINGS 14.7, 15.

6. Experiment C has run. These pages call it an open question. Selection over 30 generations left a flat contrast (-0.002, 90% CI -0.013 to +0.009). The registered verdict is uninformative: the speed control (gate 5) fails, and the preregistration interprets no contrast unless every gate passes. Gate 1 falls short only of a draft margin that was never ratified. FINDINGS 13.F.

7. The colophon. Its line "the corrected record is what these pages show" and its Version 1.0.0 citation are superseded. Cite the repository through `CITATION.cff`, which carries the current version.

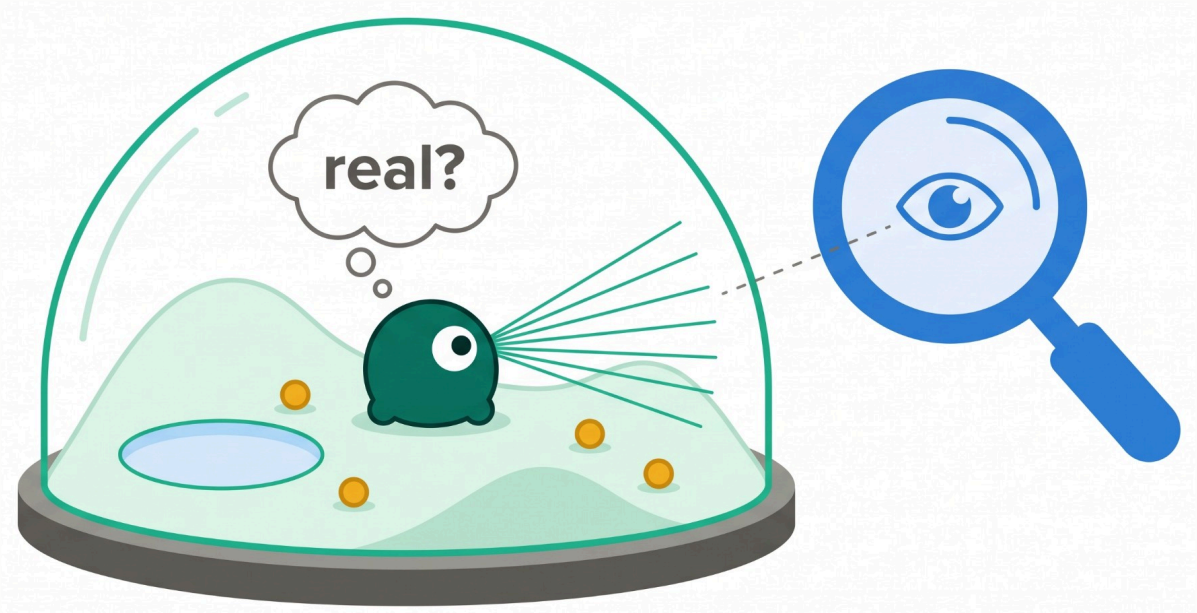
ItaSoRL

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A PRE-REGISTERED EXPERIMENT IN ARTIFICIAL LIFE

Could a tiny mind tell it was living in a simulation?

Design, control battery, pre-registered results, and open questions of the ItaSoRL research program.



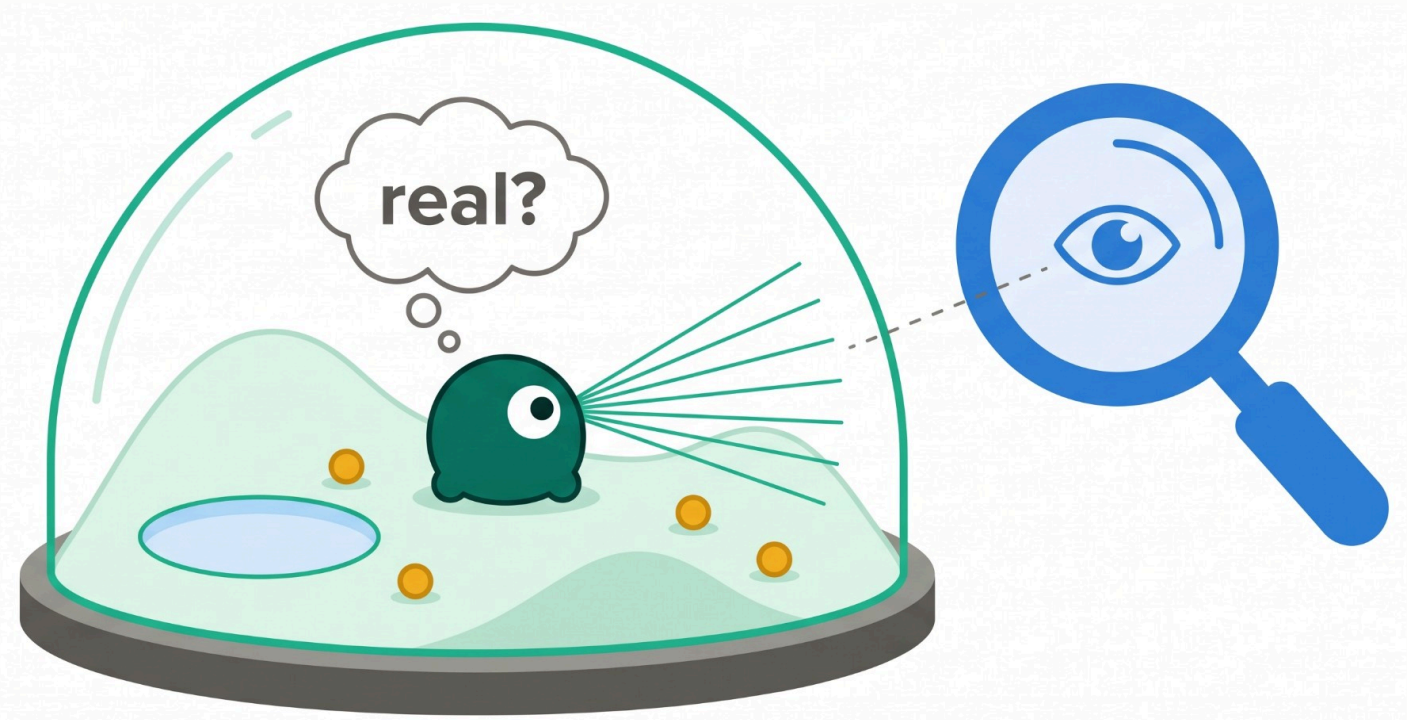
IN THIS SERIES

1. The question
2. The world
3. The agent
4. The surrogate ladder (L0-L4)
5. The matched-pair design
6. What we found
7. The evidence matrix
8. Open experiments, pre-committed rules
9. Why it matters

1 Does a minimal agent encode world identity, unasked?

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A tabula-rasa recurrent agent lives in a deterministic environment or in a surrogate copy that differs only in transition dynamics. Read-only linear probes, fitted post hoc, test whether **world identity is incidentally encoded in the hidden state** - identity never appears in observations or reward.



The agent never sees a label saying **real** or **surrogate**. Not in its senses. Not in its reward. So if the answer shows up in its memory anyway, it got there *for free* - the agent wanted it for its own reasons. That rule has a name: **readout, not reward**.

0.99 vs 0.51

The surprise. Instruments outside catch this surrogate almost every time. The agent's memory barely registers it.

0.752

The finding. Make the surrogate a *learned-dynamics rule (L3)* the agent must survive, and it quietly learns which world it lives in. Nobody asked it to.

0.65

The decision threshold. Written down *before* the decisive runs. 0.50 means guessing. Every claim on these pages faces this threshold.

THIS SERIES

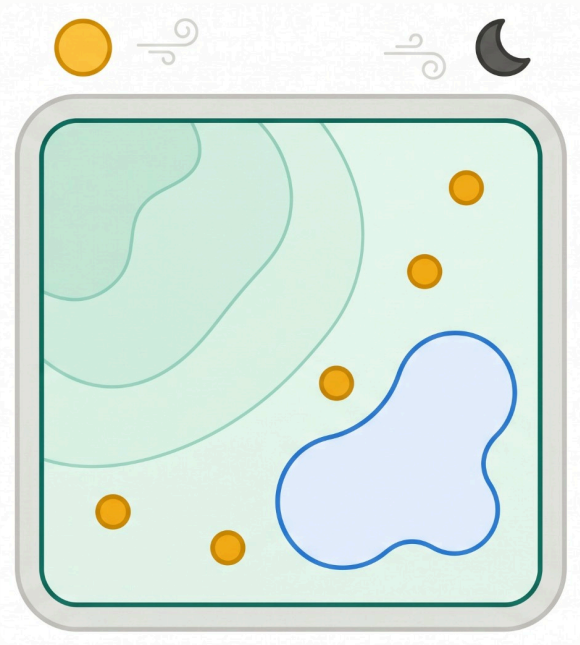
2 · The world: a fully specified dynamical system
3 · The agent: a small recurrent brain
4 · The ladder of surrogates (L0-L4)
5 · The matched-pair design that keeps it fair
6 · What the runs found
7 · The full test board
8 · Still on the bench: what remains
9 · Why it matters

THE STAGE

2 A fully specified world, copyable bit for bit

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The environment is a small, fully specified dynamical system: seeded RNG, exact update rules, bit-reproducible state. Exactness is what makes a **bit-identical control copy** possible - and with it, a falsifiable detection experiment.



WHAT THE AGENT FEELS - 146 NUMBERS, EVERY MOMENT



24 rays × 5 readings: distance, rough color, motion

smell · 12 - switched off (always zeros)

Body sense: speed, heading, energy, hydration, temperature, slope, daylight. No camera, no words, no labels. **The packet's shape never changes between real and surrogate worlds.**

WHAT IT CAN DO - 5 NUMBERS

push forward · **turn** · **eat** · **drink** · **emit scent**

Eating near food restores energy; moving and staying warm burn it. Ignore food long enough and it **dies** - in survival runs, quickly.

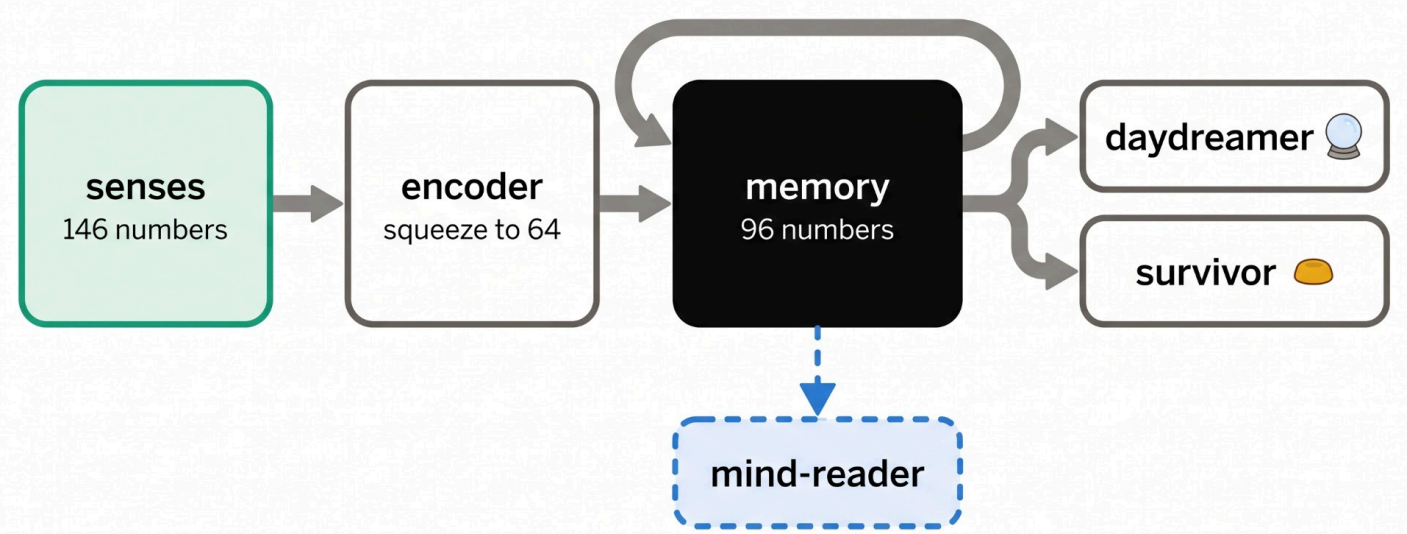
THE ONE LINE OF PHYSICS EVERY SURROGATE TOUCHES

$$v \leftarrow (1 - \mathrm{drag}\cdot dt)\cdot v + a\cdot dt$$

Speed is slowed by the medium (the **drag**) and pushed by effort. **Every surrogate world differs from the real one only inside this line.** Same senses. Same rewards. The difference lives in the physics, or nowhere.

3 A tabula-rasa recurrent agent

No pretraining, no external data. A 146-dim observation is encoded to 64 and integrated into a **~96-dim recurrent state h_t** - the only substrate in which world identity could be represented.



— TWO REGIMES, ONE BRAIN

Predictive agent (Experiment B): trained *only* to guess its next sensation. No survival game, no homework about worlds.

Survival agent (Experiment B-v2/v3): same brain shape, harsher life - it must find food or die, while the physics under its feet is base or surrogate. Survival is the *only* pressure.

— THE RULE THAT KEEPS IT SCIENCE

The probe is a **straight-line classifier fitted after training**. It never sends a single gradient back into the agent; the agent is **never graded on it**. So if the probe finds "which world am I in?" inside h_t, the agent put it there **for its own reasons**.

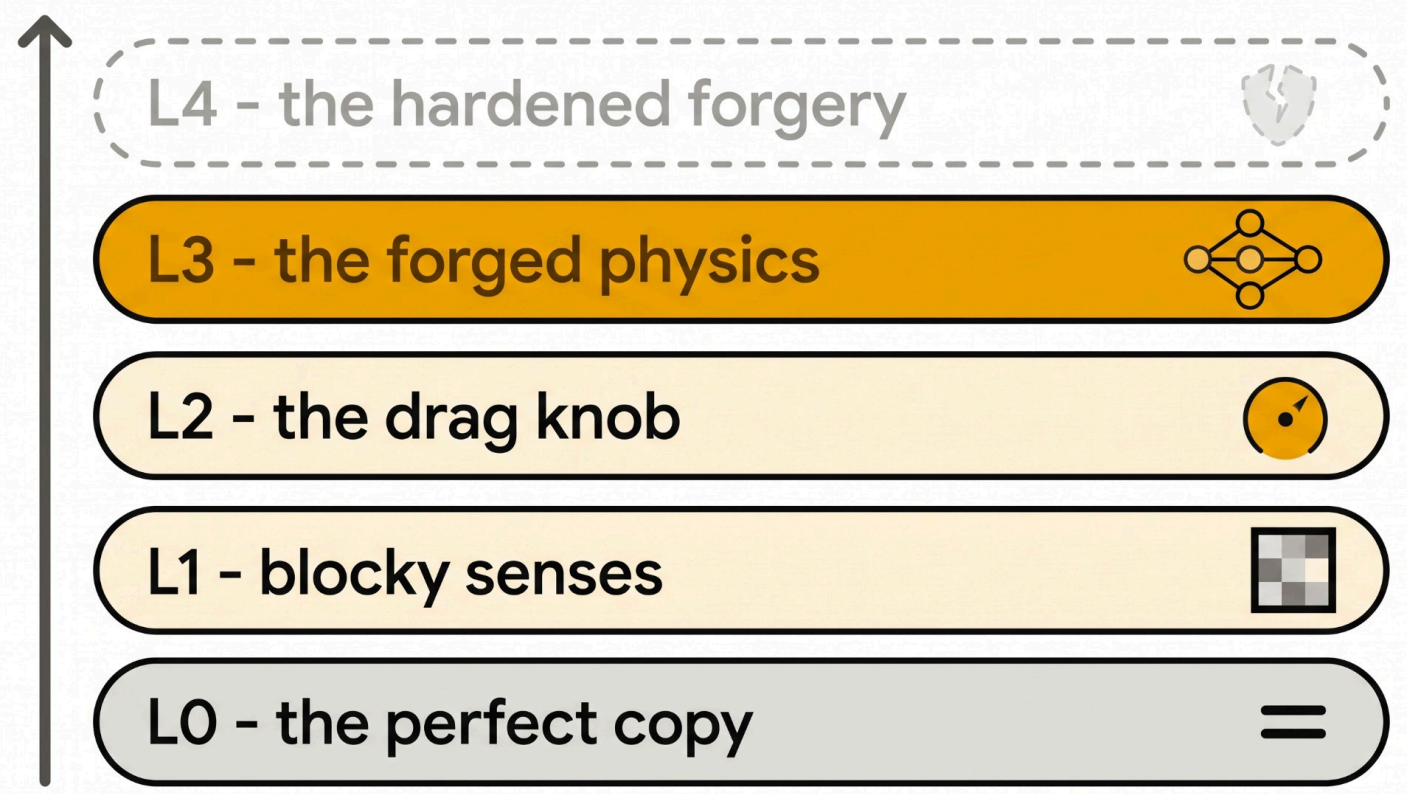
read-only | **readout, not reward**

— THE ORDER OF OPERATIONS - WHY NOTHING CAN LEAK

1. **Live & learn.** The agent trains inside its world - base or surrogate physics - knowing nothing about the experiment.
2. **Freeze.** Training ends. The brain is fixed; nothing about it changes from here on.
3. **Read.** Only now is the probe fitted - on the frozen memory, after the fact. Too late to have taught the agent anything.

The surrogate ladder (L0-L4)

All surrogates share the observation interface and reward function exactly; they differ only in transition dynamics. The ladder grades them from a bit-identical copy (L0) to a learned-dynamics surrogate (L3) and an adversarial one (L4, future).



L0 & L1 · EASY RUNGS

L0 is the trick question: a bit-identical copy, nothing changed. The probe **must** read a chance on it - it does (~0.50). **L1** snaps vision to a coarse grid - a sanity rung on the way up.

L2 · TURN THE KNOB

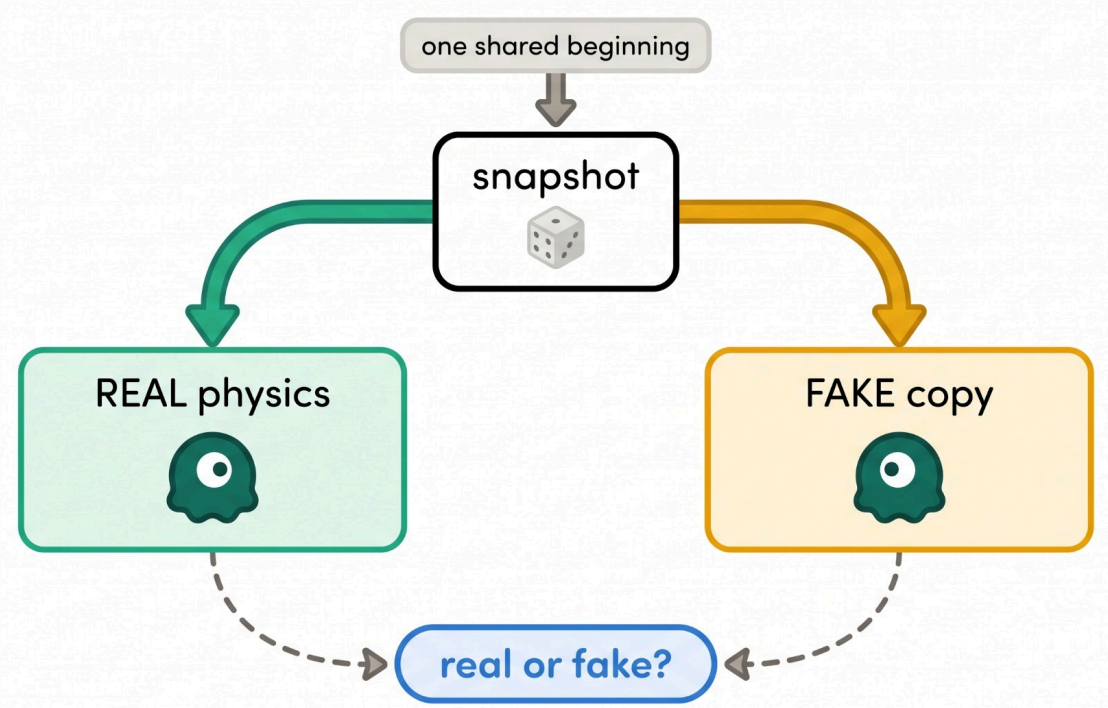
Nudge the **drag** number itself - a real, physical difference, like swimming in water vs syrup. Objectively there: an outside oracle reads it at **~0.99**.

L3 · FORGE THE RULE

A tiny network **imitates** real motion; only its small errors give it away. Calibrated first: an outside oracle must read it at **0.85-0.95** (measured: **0.928**) - clearly detectable, not trivially so.

5 Matched pairs and the control battery

Matched pairs: snapshot the full world state including RNG, then fork into base vs surrogate dynamics. Any downstream difference is **causally attributable to the dynamics switch alone**.



The snapshot copies **everything**. So the changed rule is the **only possible cause** of any difference between the twins. And the perfect-copy twins are the built-in tripwire: on them the probe **must** score a chance, or the whole pipeline is broken. It scores ~0.50. ✓

THE SCORE - AUROC



THE FAIRNESS BATTERY

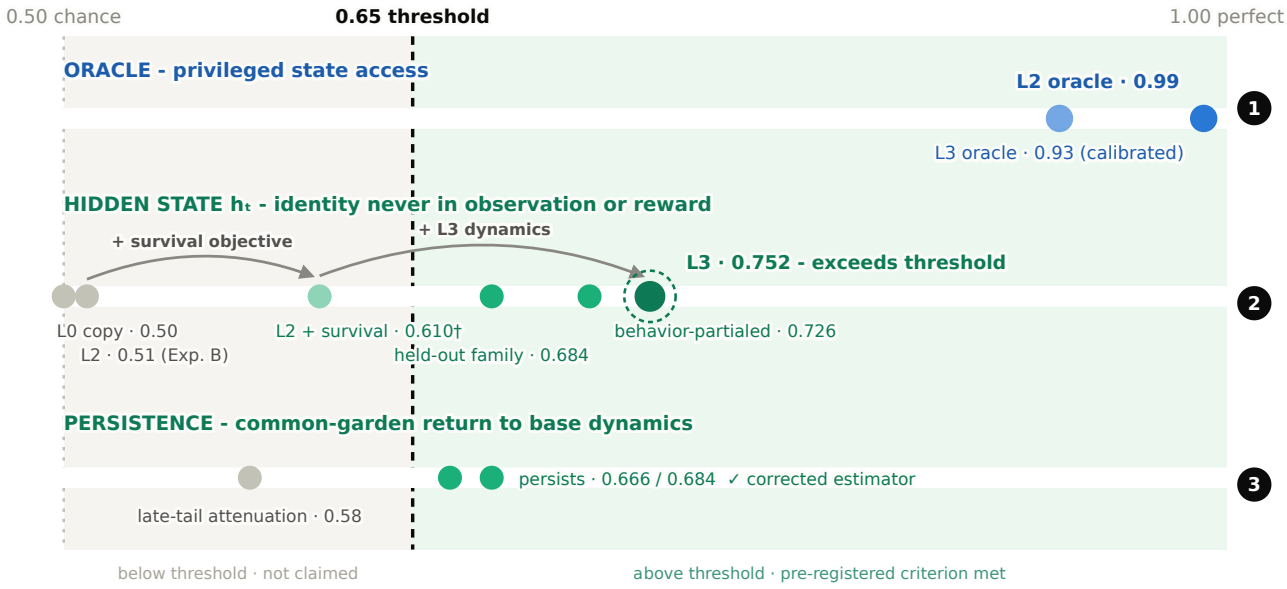
- ✓ perfect copy reads a chance - or the test is void
- ✓ reward & episode length can't leak the answer
- ✓ shuffled labels read chance - no probe magic
- ✓ behavior subtracted (speed, energy, food) to isolate what the memory *knows* from how the body *moves*
- ✓ twins never split across the probe's train/test folds

THE RESULTS

6 Externally detectable ≠ internally encoded

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Oracle detectability and hidden-state encoding dissociate - until the surrogate is a learned dynamics model under a survival objective. All values AUROC; **0.65 threshold pre-registered** before the decisive runs. Panels ①-③.



① **The surprise.** The hand-tuned L2 surrogate is blindingly obvious outside (0.99) yet a near coin-flip inside (0.51). Even under survival pressure it reaches only 0.610† - below the threshold. **Detectable is not the same as noticed.**

② **The flip.** Follow the arrows: add survival stakes, then make the surrogate a *learned-dynamics rule (L3)* - and the memory crosses the threshold at **0.752**. Still there with behavior subtracted (0.726); even reads a *different* surrogate family (0.684). Never rewarded for any of it.

③ **The memory.** Move that agent back to real physics: the signal **persists just over the threshold** (0.666/0.684) but fades late in the tail - a **modest kept belief**, not a strong stored tag. (First scoring had a bug; this is the corrected re-score.)

† 0.610 sits below the 0.65 threshold but measurably above its twin controls (~+0.10) - a small survival-specific trace, not nothing. All numbers are AUROC from committed, re-checked artifacts; the decision threshold and rules were frozen in writing before the decisive runs.

The evidence matrix

Surrogate rungs × readout regimes. Filled cells trace to committed artifacts; empty cells are explicitly not-run. All values AUROC against the pre-registered **0.65** threshold.

surrogate	oracle privileged state access	predictive Exp. B · self-supervised	survival Exp. B-v2/v3 · RL objective	evolution Exp. C · generations
L0 · bit-identical negative control	identical by construction no difference exists to read	**0.50** ✓ reads chance, as it must	**0.500** ✓ exact floor on re-score	gate must read chance or the run is void
L1 · quantized obs. sanity check	—	≈ chance nothing noticed inside	not run	—
L2 · coefficient perturbation hand-tuned	**0.99** blindingly obvious outside	**0.51** near chance inside	**0.610†** below the threshold; ≈ +0.10 over twin controls	—
L3 · learned dynamics primary test	**0.93** calibrated into 0.85-0.95 first	not run	**0.752** 0.726 behavior removed · 0.684 new recipe · keeps telling 0.666/0.684 back in real physics	queued the amplification question
L4 · adversarial surrogate future work	future work not yet built - the surrogate that fights back is the next rung of the ladder			

† small survival-specific trace: real, but below the decision threshold. A behavior-only reference (system identification) reads **0.596** - treated as a lower-bound reference, not a ceiling. Empty cells are exactly that: not run, and said so.

THE ROAD AHEAD

8 Open experiments, pre-committed rules

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Six pre-specified experiments. Decision thresholds, controls, and outcome routing (positive / negative / **uninformative**) are fixed in the preregistrations before data exist.

NEXT UP

1 · EVOLUTION - THE AMPLIFICATION QUESTION — QUEUED

One lifetime of learning built a **modest** world-detector (0.752). Experiment C asks whether **selection across generations** can amplify it into a strong, heritable one. Its first pilot was invalidated by two measurement bugs - found in-house, disclosed before the re-run's outcome was known. The re-run only counts if every gate passes:

gate 1 · surrogate certifiably exploitable → **gate 2 · both arms behave** → **gate 3 · perfect copy reads chance**

all gates pass → **amplified** or **not amplified** · any gate fails → **uninformative** - never a softer yes

DESIGNED

2 · THE DOSE-RESPONSE CURVE — PENDING

How strong must the surrogate dynamics be before the mind notices? Sweep the lie from imperceptible to obvious and map where noticing begins - the curve that sets how hard evolution's task really is.

OPEN QUESTIONS

3 · BREADTH OF THE BELIEF — OPEN

One *unseen* surrogate family read at 0.684 is a hint, not a law. Many recipes, mixed physics, harder transfers: how general is the "which world?" belief?

4 · THE MIRROR DIRECTION — OPEN

The kept-belief result is demonstrated in one direction of the swap. The mirror direction is **not demonstrated** - and gets claimed exactly that way until a dedicated run says otherwise.

THE HORIZON

5 · L4 - THE SURROGATE THAT FIGHTS BACK — FUTURE

Train the forgery *against* the detector - an arms race between hiding and noticing. Does the inside signal survive a surrogate optimized to be unnoticeable?

6 · FROM READING TO USING — FUTURE

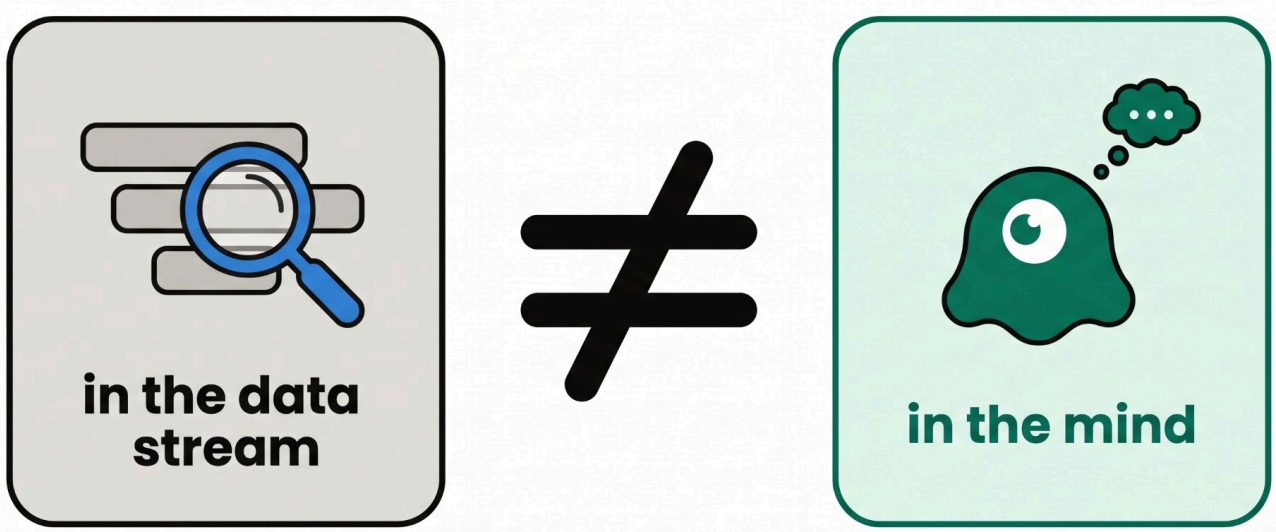
A probe proves the information is *present*; it does not prove the agent *uses* it. The causal test: intervene on the internal signal and watch whether behavior changes - the difference between a mind that could know and a mind that acts on knowing.

THE POINT

9 In the data stream ≠ in the representation

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Oracle-detectable structure in the observation process is not automatically represented in the agent's hidden state. **Encoding tracks the training objective** - representation is objective-relative, and here that is a measurement, not a slogan.



WORLD MODELS & RL

An agent that predicts its sensations - or even fights to survive - is **not** guaranteed to build an explicit "what kind of world is this?" variable. Objectives shape representations more than environments do.

INTERPRETABILITY

A probe reading information out of activations doesn't mean the system *uses* it - and information being *absent* from probes under one objective can appear under another. Probe results are objective-relative.

AI SAFETY SUBTEXT

Instrumental pressure (stay alive, reduce surprise) does not automatically produce the latent variables humans care about for oversight - but the *right kind* of pressure produced one here, unasked.

WHAT'S NEXT

Evolution. Within one lifetime, learning built a modest world-detector. Experiment C asks whether **Darwinian selection across generations** can *amplify* it into a strong, heritable one. Its first pilot was invalidated by two measurement bugs (found in an internal audit, disclosed before the outcome of the re-run was known); the gated, pre-registered re-run is queued.

The hardened forgery (L4) - a surrogate trained to resist detection - is future work.

THE HONEST CAVEAT

Twice, this project found bugs in its own scoring, said so out loud, marked the affected conclusions *provisional before knowing how re-scoring would land*, and let frozen, pre-registered rules re-decide. One verdict flipped. That is what the numbers on these pages are worth - and why.

pre-registered bars
committed artifacts
223 automated checks

COLOPHON

About this series

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RESEARCH EDITION

Everything on these pages - code, worlds, pre-registrations, findings, audit trail, and the automated checks that guard them - is open source in one repository. This page is how to find it, verify it, and cite it.

THE PROJECT

ItaSoRL - Is This A Simulation Or Real Life
Levy Tate · ORCID 0009-0009-1337-0709

A tabula-rasa artificial-life system asking whether a from-scratch digital organism comes to encode that its world is a generative surrogate rather than authentic dynamics - with detection **read out, never rewarded**. Version 1.0.0 · MIT license.

A research paper is in preparation; its title is not yet final.

REPOSITORY

`github.com/iLevyTate/ITASORL`

Key documents:
`docs/FINDINGS.md` - the full results record
`docs/PREREGISTRATION_*.md` - frozen rules & bars
`docs/AUDIT_2026-07.md` - the disclosed corrections
`scripts/` · `tests/` - runners & 223 automated checks

HOW TO CITE (SOFTWARE)

Until the paper is published, cite the repository itself - the entry below follows its committed `CITATION.cff`:

Tate, L. (2026). *ITASORL: Incidental Detection of Surrogate Worlds by a From-Scratch Digital Organism* (Version 1.0.0) [Computer software]. https://github.com/iLevyTate/ITASORL

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  title   = {ITASORL: Incidental Detection of Surrogate Worlds by a From-Scratch Digital Organism},
  year    = {2026},
  version = {1.0.0},
  license = {MIT},
  url     = {https://github.com/iLevyTate/ITASORL}
}
```

WHERE THE NUMBERS COME FROM

Every AUROC in this series is read from **committed result artifacts** in the repository - none are estimates or recollections. The 0.65 decision threshold and all decision rules were frozen in pre-registration *before* the decisive runs. Two scoring bugs found in internal audit were disclosed before their re-runs resolved; the corrected record is what these pages show.

THIS SERIES

Nine pages plus cover and this colophon · generated July 2026 from the corrected record. Illustrations generated by the author; the results chart and test board are rendered directly from committed numbers. Color code throughout: **teal = real world**, **amber = the surrogate**, **blue = the probe**.